

Review

Techniques for Measuring the Fluctuation of Residual Lower Limb Volume in Clinical Practices: A Systematic Review of the Past Four Decades

Mohd Tajularif Ibrahim ^{1,*}, Nur Afifah Hashim ^{1,*}, Nasrul Anuar Abd Razak ¹, Noor Azuan Abu Osman ¹, Hossein Gholizadeh ² and Suryani Dyah Astuti ³

¹ Department of Biomedical Engineering, University of Malaya, Kuala Lumpur 50603, Malaysia; tajularif99@gmail.com (M.T.I.); nasrul.anuar@um.edu.my (N.A.A.R.); azuan@um.edu.my (N.A.A.O.)

² Ottawa Hospital Research Institute (OHRI), Ottawa University, Ottawa, ON K1N 6N5, Canada; gholizadeh87@yahoo.com

³ Department of Physics, Faculty of Science and Technology, Airlangga University, Surabaya 60115, Indonesia; suryanidyah@fst.unair.ac.id

* Correspondence: afiqahhashim@um.edu.my

Abstract: Increased pressure and shear stress distributions at the limb–socket interface are hypothesised to result from changes in the residual limb shape and volume, which can cause socket fit difficulties. Accurate residual limb volume measurements may aid clinicians in developing strategies to accommodate volume fluctuations. This review primarily aims to analyse the techniques available for measuring the residual lower limb volume that may be used in clinical settings, as documented in the works published over the previous four decades. A comprehensive search of articles in PubMed, ScienceDirect, Web of Science, and Google Scholar identified 904 articles, and further analysis resulted in only 39 articles being chosen to be analysed. Based on the findings, there are nine techniques available to measure the residual limb volume: water displacement, anthropometric measurement, contact probes, optical scanning, spiral X-ray computed tomography (SXCT), magnetic resonance imaging (MRI), ultrasound, laser scanning, and bioimpedance. Considering the variety of techniques for determining residual limb volume, it is critical to choose the ones that best suit clinicians' objectives, and each technique has potential sources of error that should be avoided by taking precautionary action. A comprehensive study of the measurement techniques is needed since researchers have developed and extensively utilised many new measuring devices, especially handheld 3D laser scanners.

Keywords: biomechanics; biomechanical engineering; lower limb; medical devices; orthopaedic bioengineering; residual limb volume; prosthesis fitting; prosthetics



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1. Introduction

The residual limb normally changes in shape and volume during the post-operative recovery period, which is between 12 and 18 months following amputation surgery [1–4]. Even after 18 months following the amputation surgery (the mature phase), the residual limb continues to experience daily volume fluctuation. Several factors may contribute to this residual limb volume fluctuation, such as comorbidities [4,5], daily activity [6–8], and prosthesis suspension [9–11]. This daily volume fluctuation varies according to the individual [5,12] and may contribute to difficulties in conducting the prosthetic socket fitting. In some designs of prosthetic sockets such as total surface bearing (TSB) prosthetic sockets, proper prosthetic socket fitting is determined by the equal distribution of the interface pressure and shear stress within the prosthetic socket [13–16]. According to Roy et al. [17] and Ahmadizadeh et al. [18], changes in the residual limb volume have an impact on the socket fit that may result in adverse modifications to the prosthetic socket

interface. These modifications increase the harmful shear stress and cause skin damage to the residual limb. A study by Pezzin et al. [19] reported that nearly one-third of amputees expressed dissatisfaction with their prostheses, and according to Afzal et al. [20], 75% of the amputees experienced skin problems including irritation, ulcers, the inclusion of cysts, and blisters. During the prosthetic fitting process, the prosthetic socket must be altered regularly since the residual limb shape and volume are constantly changing. This condition develops because of oedema [2,21–23], the presence of discreate fluid [24], and muscular build-up or atrophy [21]. Monitoring and managing the residual limb volume in the lower extremity is essential because physicians and prosthetists rely on these factors to determine the optimal time and the best type of prosthesis for the amputees. Given the regularity of residual limb volume measurements and their significance in prosthetic prescription, it is crucial for clinicians and prosthetists to have access to techniques that enable the evaluation of the lower extremity residual limb volume.

Researchers are interested in examining how much residual limb volume fluctuation exists and how the amount of fluctuation can be measured through various techniques including water displacement, casting + water displacement, anthropometric measurement, contact probes, spiral X-ray computed tomography (SXCT), magnetic resonance imaging (MRI), ultrasound, optical scanning, laser scanning, and bioimpedance, as reported by Sanders and Fatone [4]. Although numerous studies have been carried out on residual limb volume fluctuation using a variety of techniques, there has been a dearth of scholars who have conducted systematic reviews of the existing studies. There are only two studies conducted by Sanders and Fatone [4] and Armitage et al. [25] who assessed and analysed the various techniques for measuring the residual limb volume in patients with lower limb amputations. Sanders and Fatone [4] identified a vast array of measurement techniques, ranging from simple techniques (e.g., water displacement and anthropometric measurement) to complex ones such as laser scanning, SXCT, MRI, and bioimpedance. The exhaustive analysis discussed the benefits and drawbacks of each measurement technique, as well as their associated errors in volume change. Then, Armitage et al. [25] evaluated the reliability and validity of some measuring techniques discussed by Sanders and Fatone [4] and finally provided and extracted the quantitative data on the validity and reliability of the techniques (i.e., correlation coefficients). Armitage et al. [25] also assessed the quality of the measurement techniques to see if the research findings could be trusted and if the present measurement techniques for residual limb volume were valid and reliable. Since the last systematic review on residual limb volume measurement techniques was published in 2011 by Sanders and Fatone [4], and the 2019 publication by Armitage et al. [25] focused primarily on the reliability and validity of the measurement techniques, researchers believe that a new systematic review is required. Although both articles were published in 2011 and 2019, Sanders and Fatone [4] only conducted the studies until 2009, while Armitage et al. [25] only conducted research through the 2016 publication. Over the past ten years, it appears that some researchers have made advancements to the current techniques used to evaluate the residual limb volume, making them simpler, more precise, and more user-friendly, especially for clinicians to use in clinical settings. Additionally, a comprehensive study of the measurement techniques is needed since new measuring devices such as handheld 3D laser scanners were developed and are extensively utilised by researchers.

The review is guided by the following central research questions: (1) What techniques are available for measuring the residual limb volume fluctuation? (2) What is the procedure for performing the techniques? This study attempts to close the knowledge gap by conducting a systematic review of previous related studies to gain a better understanding of the available techniques for measuring the residual lower limb volume. It is now clear that there is a fluctuation in the residual limb volume during the period of prosthetic usage and that there are techniques that can be used to measure this fluctuation. Healthcare professionals will benefit the most from this review's information because it will enable them to comprehend the potential applications of measuring techniques in clinical settings.

Additionally, it will be useful to academics, researchers, prosthetists, and even prosthesis users who are interested in the subject. This review can potentially close the information gap among these interested parties regarding the residual lower limb volume measuring techniques, which seem to become more practical and easier to apply from year to year. Moreover, several measuring techniques have recently seen a lot of implementations in clinical studies compared to the years before 2009, as reported by Sanders and Fatone [4]. From this collection of works, the interested parties will discover more about residual limb volume and design an adaptation approach that suits their own needs, capabilities, and interests in specifically regulating residual limb volume fluctuation.

2. Methods

2.1. The Review Protocols

In this systematic review, the authors followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) [25] guidelines, which are the gold standard in the field [26]. The authors concentrated their efforts on seeking any related terms for the main keywords used, which included residual limb, fluctuation, volume, technologies, techniques, and measurement (Appendix A). Search functions such as Boolean operators, phrase searching, truncation, wild cards, and field code function were used to combine the enriched keywords. This systematic research of the literature was performed through all four main databases, namely PubMed, ScienceDirect, Web of Science, and Google Scholar.

2.2. Selection of Studies

The inclusion criteria include (1) type of literature: journal articles were selected as the primary kind of literature as they provided primary data that were essential for the study; (2) the language: written in English only; (3) the year of publication: the review should cover a 47-year period (publications that were published between 1975 and 2022 were included); and (4) publications that describe and perform any residual lower limb volume measurement techniques by either measuring the residual limb volume as a whole or only measuring the residual limb volume changes, providing a residual lower limb assessment, and only evaluating the residual limb volume change in individuals with lower limb amputation.

The exclusion criteria were as follows: (1) articles without abstracts; (2) articles written in a language other than English; and (3) studies that do not provide any assessment of the residual lower limb and do not use real/model residual lower limbs to measure the volume. Five reviewers who looked over all the abstracts reached an agreement on which ones should be included or excluded.

2.3. Data Extraction and Quality Assessment

At this stage, the authors read the rest of the articles, particularly the abstract, methodology, and results sections. Subsequently, the authors gathered all the techniques reported in the articles specifically from their methodology sections, and all the data concerning the participants' characteristics and details as well as the procedure were extracted.

The authors developed one quality rating scale to assess the quality of all the shortlisted articles. This quality rating scale was adapted from the Quality Appraisal of Diagnostic Reliability Accuracy, Quality Assessment of Diagnostic Accuracy Studies (QUADAS-2), and the study conducted by Armitage et al. In the developed quality rating scale, four main elements were evaluated: (1) assessment of subject selection and their details; (2) assessment of methodology to determine whether the methodology had enough details (describing the procedure for measuring the residual limb) to allow for replications; (3) assessment of reference standard; and (4) assessment of results (Appendix B). This quality rating scale consisted of 13 questions that had to be answered with scores of 'Yes = 1', 'No = 0', and 'Unclear = 0.5', and all the questions had the same weight. The authors then summed up all the scores. The publications were considered high quality if

they fulfilled more than 9 items. Moderate-quality publications were those that fulfilled 5 to 8 items. Finally, the publications were considered low quality if they only fulfilled less than 4 items.

2.4. Data Synthesis and Analysis

The articles with the same techniques were gathered for the purpose of data synthesis. The authors focused the data synthesis on the five main themes: (1) subject selection characteristic—the type of amputation and the number of subjects; (2) comparison between techniques used; (3) procedure used to perform the test—general procedure, subject, and limb positions during measurement; (4) possible sources of error and precautionary steps taken; and (5) general publication details, e.g., latest publications, most popular, and easy-to-apply techniques. Additionally, data analysis based on the comparison of the techniques' costs, times/durations, accuracy and precision, setup and availability, and applications were carried out. In the final section of the analysis, the authors assessed the articles using reliability and validity as focal points. The study evaluation criteria for reliability and validity analysis are provided in Appendix C. The analysis was conducted to ascertain whether the publications conducted tests of reliability and/or validity only. However, this scope of analysis did not encompass an in-depth examination, such as a comprehensive statistical assessment of reliability or validity.

3. Results

3.1. Studies' Eligibility

The methodology described in the preceding section was applied, resulting in an initial collection of 904 articles (refer to Figure 1). Subsequently, 480 duplicate references were eradicated using the Mendeley software 1.19.8. Another 399 references were excluded during the screening phase, adhering to the predetermined inclusion and exclusion criteria. This left a total of 81 articles for further consideration. A team of reviewers undertook the task of manually reviewing these 81 references. During this process, 42 articles were excluded due to insufficient methodologies and inadequately presented procedures. This elimination was based on the manual screening, during which the authors concentrated on the abstract, methodology, and results sections. The remaining 39 papers underwent an evaluation of both their overall quality and the process of data extraction.

3.2. Data Extraction

In the data extraction process, all 39 articles were gathered under the same techniques used. Table 1 shows the nine techniques that can be used to measure the residual lower limb volume fluctuation. These techniques include water displacement, anthropometric measurement, contact probes, optical scanning, spiral X-ray computed tomography (SXCT), magnetic resonance imaging (MRI), ultrasound, laser scanning, and bioimpedance. Of the nine techniques, the highest number of publications is focused on the bioimpedance technique ($n = 15$) followed by laser scanning ($n = 9$), optical scanning ($n = 9$), water displacement ($n = 8$), anthropometric measurement ($n = 5$), contact probes ($n = 3$), ultrasound and SXCT ($n = 2$), and the lowest number of publications is focused on MRI ($n = 1$).

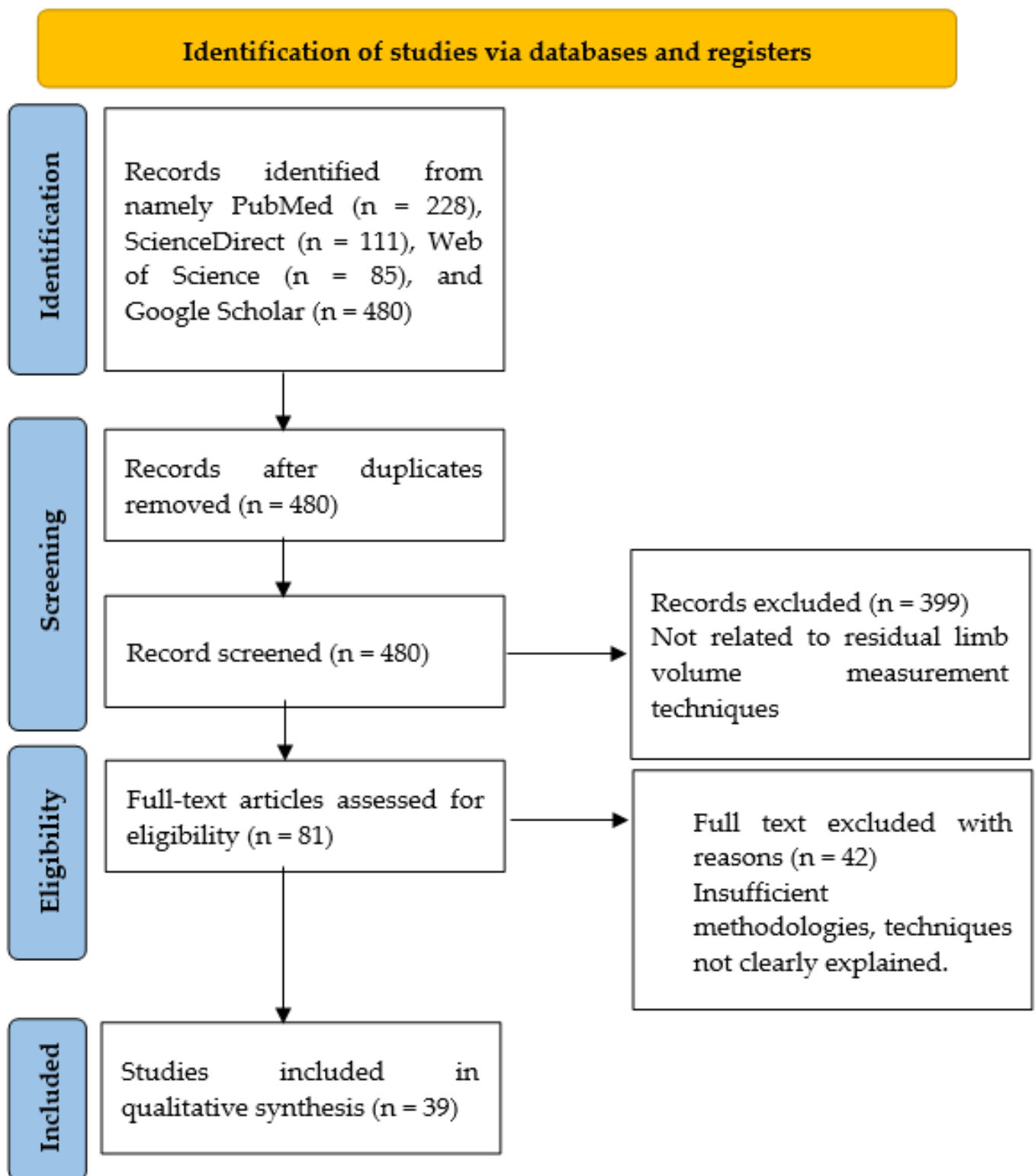


Figure 1. PRISMA 2009 flow diagram. The diagram shows the number of publications identified and selected for the review.

3.3. Publication Quality Assessment

All 39 publications were analysed for their quality assessment, and the results are summarised in Table 2. A total of 22 studies were classified as high-quality articles, 14 publications were classified as moderate-quality articles, and 3 publications were classified as low-quality articles. Most of the publications were classified as high-quality articles

because they have a clear methodology that allow for replications by other researchers. On the other hand, three publications were classified as low-quality articles because they do not have a clear data collection procedure and they do not discuss the analysis of their findings in detail. Despite the low-quality rankings assigned to these three articles, the authors chose to analyse the findings because the publications were successful in showcasing that the employed techniques could quantify the residual limb volume.

Table 1. Residual limb volume measuring techniques.

Residual Limb Volume Measuring Techniques	Authors
Water Displacement	Fernie et al. [26]; Thomas W. Starr [27]; Golbranson et al. [21]; Krouskop et al. [28]; Smith et al. [29]; Sven Johansson and Öberg [30]; Boonhong [31]; de Boer-Wilzing et al. [32]
Anthropometric Measurement	Golbranson et al. [21]; Krouskop et al. [28]; Boonhong [31]; de Boer-Wilzing et al. [32]; Tantua et al. [33]
Contact Probes	Krouskop et al. [28]; Vannah et al. [34]; McGarry et al. [35]
Optical Scanning	Schreiner and Sanders [36]; Smith et al. [29]; Vannier et al. [37]; Johansson and Öberg [30]; Zachariah et al. [5]; Sanders and Lee [38]; de Boer-Wilzing et al. [32]; Dickinson et al. [39]; Kofman et al. [40]
Spiral X-Ray Computer Tomography (SXCT)	Smith et al. [29]; Commean et al. [1]
Magnetic Resonance Imaging (MRI)	Buis et al. [41]
Ultrasound	He et al. [42]; Singh et al. [24]
Laser Scanning	Fernie et al. [43]; Öberg et al. [44]; Johansson and Öberg [30]; de Boer-Wilzing et al. [32]; Tantua et al. [33]; Dickinson et al. [39]; Seminati et al. [45]; Kofman et al. [40]; Paternò et al. [46]
Bioimpedance	Miyatani et al. [47]; Sanders et al. [48]; Sanders et al. [12]; Sanders et al. [49]; Sanders et al. [50]; Sanders et al. [51]; Sanders et al. [52]; Sanders et al. [6]; Sanders et al. [53]; Sanders et al. [54]; Sanders et al. [7]; Youngblood et al. [8]; Hinrichs et al. [55]; Youngblood et al. [56]; Larsen et al. [57]

3.4. Data Synthesis

Based on the findings in the publications from the past 10 years (2011–2021) (Figure 2), there were a high number of publications for the bioimpedance ($n = 12$) and laser scanning ($n = 6$) techniques compared to the other techniques, namely water displacement ($n = 1$), anthropometric measurement ($n = 2$), and optical scanning ($n = 3$). Also, there were no publications on the contact probe, SXCT, MRI, or ultrasound techniques for the past 10 years. According to the study by the authors, Sanders wrote a significant number of articles on bioimpedance techniques (10 out of 15 publications), even though laser scanning techniques garnered greater attention from different scholars. There were more scholarly publications, which increased the variety of study designs that were used in laser scanning techniques. In addition, from the analysis, the authors also discovered that the water displacement, anthropometric measurement, optical scanning, laser scanning, and bioimpedance techniques were still practical to use, as all the techniques were still being used by many researchers for the past three consecutive decades.

Table 2. Quality assessments of the publications.

Study	Question Number													* Rating Score
	1	2	3	4	5	6	7	8	9	10	11	12	13	
Fernie et al. [26]	✓	X	X	X	✓	X	X	X	X	✓	?	?	?	4.5 (c)
Thomas W. Starr [27]	✓	X	X	X	✓	?	X	✓	X	✓	?	?	X	5.5 (b)
Fernie et al. [43]	✓	X	X	X	✓	X	X	X	X	✓	?	?	X	4 (c)
Golbranson et al. [21]	✓	✓	X	X	✓	✓	✓	✓	✓	✓	✓	✓	✓	11 (a)
Krouskop et al. [28]	✓	X	X	X	✓	✓	X	X	✓	✓	✓	?	X	6.5 (b)
Öberg et al. [44]	✓	✓	✓	X	✓	✓	X	X	X	X	✓	?	X	6.5 (b)
Schreiner et al. [36]	✓	✓	X	X	?	?	✓	✓	X	✓	✓	?	X	6.5 (b)
Smith et al. [29]	✓	✓	?	✓	✓	✓	?	✓	✓	✓	✓	✓	✓	12 (a)
Vannier et al. [58]	✓	X	X	X	✓	✓	X	✓	✓	✓	✓	✓	✓	9 (a)
He et al. [42]	✓	X	X	X	✓	✓	X	X	X	✓	✓	X	X	5 (b)
Vannah et al. [34]	✓	X	X	X	?	✓	X	✓	X	✓	✓	✓	✓	7.5 (b)
Commean et al. [1]	✓	✓	X	✓	✓	✓	✓	✓	X	X	✓	✓	✓	10 (a)
Johansson and Oberg [30]	✓	X	X	X	✓	✓	X	✓	✓	✓	✓	✓	✓	9 (a)
Boonhong [31]	✓	X	X	✓	✓	✓	X	✓	✓	✓	✓	✓	✓	10 (a)
Zachariah et al. [5]	✓	✓	X	✓	✓	X	✓	✓	X	X	✓	✓	✓	9 (a)
Buis et al. [41]	✓	X	X	X	✓	X	X	X	X	✓	?	X	X	3.5 (c)
Singh et al. [24]	✓	✓	✓	X	?	X	X	✓	X	X	✓	✓	✓	7.5 (b)
Sanders et al. [48]	✓	✓	✓	✓	✓	✓	✓	✓	X	X	✓	✓	✓	11 (a)
McGarry et al. [35]	✓	X	X	X	?	✓	X	✓	X	X	✓	✓	✓	6.5 (b)
Sanders and Lee, [38]	✓	X	X	X	✓	✓	X	✓	X	X	✓	?	?	6 (b)
Sanders et al. [12]	✓	✓	✓	✓	✓	✓	✓	✓	X	X	✓	X	X	9 (a)
De Boer-Wilzing et al. [32]	✓	✓	X	✓	✓	✓	X	✓	✓	✓	✓	✓	✓	11 (a)
Sanders et al. [49]	✓	✓	X	✓	✓	✓	✓	✓	X	X	✓	X	X	8 (b)
Sanders et al. [51]	✓	✓	✓	✓	✓	✓	✓	✓	X	X	✓	✓	✓	11 (a)
Sanders et al. [50]	✓	✓	✓	✓	✓	✓	✓	✓	X	X	✓	✓	✓	11 (a)
Sanders et al. [52]	✓	✓	X	✓	✓	✓	✓	✓	X	X	✓	✓	✓	10 (a)
Tantua et al. [33]	✓	✓	✓	X	✓	✓	?	?	✓	✓	✓	✓	✓	11 (a)
Sanders et al. [6]	✓	✓	✓	✓	✓	✓	✓	✓	X	X	✓	✓	✓	11 (a)
Sanders et al. [53]	✓	✓	X	✓	✓	✓	✓	X	X	X	✓	?	X	7.5 (b)
Dickinson et al. [39]	✓	X	X	X	✓	✓	X	✓	X	X	✓	✓	✓	7 (b)
Seminati et al. [45]	✓	X	X	X	✓	✓	X	✓	X	✓	✓	✓	✓	8 (b)
Sanders et al. [54]	✓	✓	X	✓	✓	✓	✓	✓	X	X	✓	✓	✓	10 (a)
Kofman et al. [40]	✓	X	X	X	✓	✓	X	✓	X	✓	✓	✓	✓	8 (b)
Sanders et al. [7]	✓	✓	X	✓	✓	✓	✓	✓	X	X	✓	✓	✓	10 (a)
Hinrichs et al. [55]	✓	✓	✓	✓	✓	✓	✓	✓	X	X	✓	✓	✓	11 (a)
Youngblood et al. [8]	✓	✓	✓	✓	✓	✓	✓	✓	X	X	✓	✓	✓	11 (a)
Larsen et al. [57]	✓	✓	X	✓	✓	✓	✓	✓	X	X	✓	✓	✓	10 (a)
Youngblood et al. [56]	✓	✓	X	✓	✓	✓	✓	✓	X	X	✓	✓	✓	10 (a)
Paternò et al. [46]	✓	✓	✓	✓	✓	✓	✓	✓	X	X	✓	✓	✓	11 (a)

* Rating Score: ✓: Yes (1); X: No (0); ?: Unclear (0.5); (a): High quality (≥ 9); (b): Moderate quality; (c): Low quality (≤ 4).

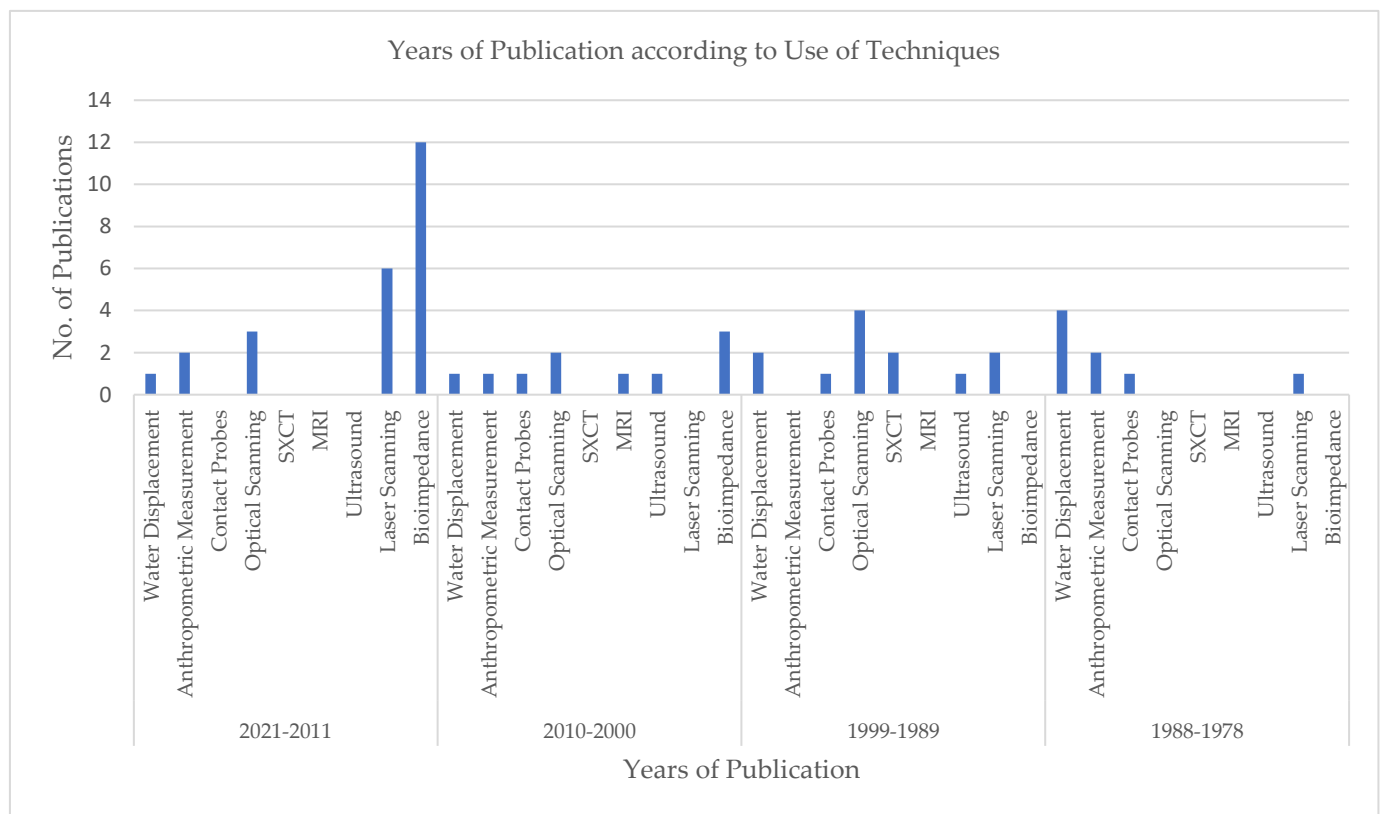


Figure 2. Number of publications on residual limb volume measuring techniques.

Table 3 provides a summary of each publication. All 39 articles were grouped according to the techniques used to measure the residual limb volume, the level of the subjects' amputations, the number of subjects, the positions of the subjects during testing, and the possible sources of error. These studies were compiled based on the year published (from the oldest year to the most recent year). It was discovered that some studies used more than one technique to determine the validity of the techniques. In terms of the subjects' amputation levels used in the studies, transtibial amputation showed a much higher number of publications, with 36 publications, compared to transfemoral amputation ($n = 5$). Other than that, some studies also used able bodies ($n = 2$) and other shapes such as abstract shapes ($n = 7$) to replace the residual limb, especially in the validity studies.

Table 3. Summary of the selected articles on residual limb volume measuring techniques. Order is based on publication year (from oldest to most recent year).

Study	Technique Used	Amputation Level (n)	Subject/Model Position during Test				Source of Possible Error				
			Supine/Prone	Seat	Stand	Walk	UC	SM	LSD	PD	R
Fernie et al. [26]	Water Immersion	Transfemoral; Other (1, 4)			✓			✓	✓	✓	
Thomas W. Starr [27]	Water Immersion	Transtibial; Other (4, 4)		✓	✓			✓	✓	✓	
Fernie et al. [43]	Laser Scanning	Transtibial; Other (1, 1)		✓				✓		✓	✓

Table 3. Cont.

Study	Technique Used	Amputation Level (n)	Subject/Model Position during Test					Source of Possible Error				
			Supine/Prone	Seat	Stand	Walk	UC	SM	LSD	PD	R	Ot (SE)
Golbranson et al. [21]	Water Immersion; Anthropometric Measurement	Transtibial (36)		✓				✓	✓	✓		
Krouskop et al. [28]	Water Immersion; Anthropometric Measurement; Contact Probes	Transfemoral (5 + 100)		✓	✓			✓	✓	✓		
Öberg et al. [44]	Laser Scanning	Transtibial (21)		✓				✓		✓	✓	
Schreiner et al. [36]	Optical Scanning	Transtibial; Other (1, 1)					✓		✓	✓	✓	
Smith et al. [29]	Water Immersion; Optical Scanning; Computer Tomography	Transtibial (10)	✓					✓	✓	✓	✓	
Vannier et al. [58]	Optical Scanning; Computer Tomography; Other	Transtibial (13)		✓				✓	✓	✓	✓	
He et al. [42]	Ultrasound	Transtibial (1)		✓						✓	✓	
Vannah et al. [34]	Contact Probes	Transtibial; Other (1, 3)					✓	✓	✓	✓	✓	
Commean et al. [1]	Computer Tomography	Transtibial (1)	✓					✓		✓	✓	
Johansson and Oberg [30]	Water Immersion; Optical Scanning; Laser Scanning; Other	Transtibial; Other (6, 3)						✓	✓	✓	✓	
Boonhong [31]	Water Immersion; Anthropometric Measurement	Transtibial (55)		✓				✓	✓	✓		
Zachariah et al. [5]	Optical Scanning	Transtibial (6)		✓		✓		✓		✓	✓	
Buis et al. [41]	Magnetic Resonance Imaging	Transtibial (1)	✓					✓		✓	✓	
Singh et al. [24]	Ultrasound	Transtibial; Transfemoral (62, 43)					✓			✓	✓	
Sanders et al. [48]	Bioimpedance	Transtibial; Other (4, 2)	✓	✓	✓				✓		✓	✓
McGarry et al. [35]	Contact Probes	Transtibial (1)					✓	✓		✓	✓	
Sanders and Lee, [38]	Optical Scanning	Transtibial (-)					✓	✓		✓	✓	
Sanders et al. [12]	Bioimpedance	Transtibial (4)		✓	✓	✓			✓		✓	✓

Table 3. Cont.

Study	Technique Used	Amputation Level (n)	Subject/Model Position during Test					Source of Possible Error				
			Supine/Prone	Seat	Stand	Walk	UC	SM	LSD	PD	R	Ot (SE)
De Boer-Wilzing et al. [32]	Water Immersion; Anthropometric Measurement; Optical Scanning; Laser Scanning, Other	Transtibial (26)					✓	✓	✓	✓	✓	
Sanders et al. [49]	Bioimpedance	Transtibial (7)		✓	✓	✓			✓		✓	✓
Sanders et al. [51]	Bioimpedance	Transtibial (30)		✓	✓	✓			✓		✓	✓
Sanders et al. [50]	Bioimpedance	Transtibial (12)		✓	✓	✓			✓		✓	✓
Sanders et al. [52]	Bioimpedance	Transtibial (19)		✓	✓	✓			✓		✓	✓
Tantua et al. [33]	Anthropometric Measurement, Laser Scanning	Transtibial (21)		✓				✓		✓	✓	
Sanders et al. [6]	Bioimpedance	Transtibial (24)		✓	✓	✓			✓		✓	✓
J. E. Sanders et al. [53]	Bioimpedance	Transtibial; Transfemoral (3, 8)	✓	✓					✓		✓	✓
Dickinson et al. [39]	Optical Scanning; Laser Scanning; Other	Transtibial (20)					✓	✓		✓	✓	
Seminati et al. [45]	Laser Scanning	Transtibial; Transfemoral (5, 5)					✓	✓		✓	✓	
Sanders et al. [54]	Bioimpedance	Transtibial (9)		✓	✓	✓			✓		✓	✓
Kofman et al. [40]	Optical Scanning; Laser Scanning	Transtibial; Other (6, 3)					✓	✓		✓	✓	
Sanders et al. [7]	Bioimpedance	Transtibial (29)		✓	✓	✓			✓		✓	✓
Hinrichs et al. [55]	Bioimpedance	Transtibial (2)		✓	✓	✓			✓		✓	✓
Youngblood et al. [8]	Bioimpedance	Transtibial (13)		✓	✓	✓			✓		✓	✓
Larsen et al. [57]	Bioimpedance	Transtibial (12)							✓		✓	✓
Youngblood et al. [56]	Bioimpedance	Transtibial (12)		✓	✓	✓			✓		✓	✓
Paternò et al. [46]	Laser Scanning	Transfemoral (24)			✓			✓		✓	✓	

UC = Unclear or Other Position; SM = Subject Movement; LSD = Limb Shape Distortion; PD = Post-Doffing Effect; R = Device Resolution; Ot (SE) = Other Sources of Error or Not Stated.

Table 4 displays each technique's capacity to capture the residual limb shape and/or volume, as well as how to measure volume fluctuation and the conditions required for volume measurement. All measurement techniques, except for anthropometric measurement, bioimpedance, and water displacement, may be used to assess the residual limb's volume and shape. The water displacement and bioimpedance techniques have distinct limitations because they can only evaluate volume, in contrast to anthropometric measurements, which only identify circumferences and require calculations to calculate volume. All measuring techniques can therefore be used to determine the residual limb volume, except for bioimpedance and SXCT, which can only detect volume change. Additionally, while the water displacement, contact probe, and bioimpedance techniques require a particular experiment setup, the anthropometric measurement, optical scanning, SXCT, MRI, ultrasound, and laser scanning techniques can be employed in clinical settings without a particular setup.

Table 4. Summary of the techniques according to ability, method, and setting needed to capture the residual limb's shape and/or volume.

Technique	Ability to Capture Shape and/or Volume			Method of Measuring the Fluctuation			Setting	
	Shape	Volume	Other Measurements	Whole Volume	Change in Volume Only	Other Measurements	Clinical Setting	Lab Setting
Water Displacement	-	✓	-	✓	-	-	-	✓
Anthropometric Measurement	-	-	✓	-	-	✓	✓	✓
Contact Probes	✓	✓	-	✓	-	-	-	✓
Optical Scanning	✓	✓	-	✓	-	-	✓	✓
SXCT	✓	✓	✓	-	✓	-	✓	-
MRI	✓	✓	✓	✓	-	-	✓	-
Ultrasound	✓	✓	✓	✓	-	-	✓	-
Laser Scanning	✓	✓	✓	✓	-	-	✓	✓
Bioimpedance	-	✓	-	-	✓	-	-	✓

On the other hand, Table 5 compares the techniques based on their costs, setup and availability, accuracy and precision, time/duration, and applications. According to the analysis, each technique has its benefits and drawbacks.

Table 5. Comparison of the techniques based on cost, time/duration, accuracy and precision, setup and availability, and applications.

Theme	Technique	Details
Cost	Water Displacement	Cost-effective—does not require the utilisation of costly equipment or technology
	Anthropometric Measurement	Cost-effective—does not require the utilisation of costly equipment, normally only use measurement tape
	Contact Probes	Expensive—more difficult to reach some clinics or patients because this technique requires proper setup and pricey equipment
	Optical Scanning	Expensive equipment—requires specialised equipment, which can be expensive as well as difficult to obtain

Table 5. Cont.

Theme	Technique	Details
	SXCT	Expensive equipment—requires highly specialised and pricey equipment
	MRI	Expensive equipment—data collection for MRI machines requires specialised, costly instruments
	Ultrasound	Low-cost equipment—more clinics and patients can use ultrasound equipment because it is less expensive than other measurement methods like contact probes and laser scanning
	Laser Scanning	Expensive equipment—specialised equipment needed for laser scanning might be expensive and difficult to obtain
	Bioimpedance	Costly—some clinics or patients might not be able to afford bioimpedance measurement tools due to their high cost and complicated setup
Time/Duration	Water Displacement	Time-consuming—using water displacement to measure residual limb volume might take some time, especially for bigger/larger limbs
	Anthropometric Measurement	Real-time measurement and fast measuring technique—only requires a clinician to use a measuring tape and measure the circumference at a specified location
	Contact Probes	Real-time measurements—enable medical professionals to monitor changes and make necessary corrections; however, data analysis takes some time
	Optical Scanning	Time-consuming—the scanning process is quick, but the data processing and analysis necessary for precise measurements can take some time
	SXCT	Time-consuming—the SXCT technique might take an extended period because it involves extensive scanning and data processing
	MRI	Time-consuming—the MRI process can take up to an hour to complete
	Ultrasound	Real-time measurements—real-time measurements are provided via ultrasound; however, data analysis takes some time
	Laser Scanning	Time-consuming—even though scanning takes a short time, the data processing and analysis needed for precise measurements can take some time
	Bioimpedance	Real-time measurements—offers real-time measurements, yet it takes some time to analyse the data
Accuracy and Precision	Water Displacement	Accurate—water displacement is an extremely accurate method to calculate the volume of the residual limb. The measurements are accurate and may be verified by repeating them several times
	Anthropometric Measurement	Limited accuracy—less precise than other techniques, particularly in individuals with limbs with uneven shape or who have oedema
	Contact Probes	High precision—very accurate and precise, enabling the detection of small changes in volume Repeat measures—several measurements may be performed from various locations on the residual limb to verify accuracy; this provides a more accurate average volume and helps to correct for any shape irregularities
	Optical Scanning	High accuracy—extremely precise measurements, with accuracy to within a tenth of a millimetre
	SXCT	Accurate and precise measurements—SXCT can yield measurements of the residual limb volume that are extremely accurate and precise, with accuracy to within a fraction of a millimetre
	MRI	High-quality 3D images—high-quality 3D scans from MRI provide extensive information regarding the volume of the remaining limb
	Ultrasound	Limited precision—the precision of ultrasound measurements may not be as high as that of other techniques like contact probes or laser scanning

Table 5. Cont.

Theme	Technique	Details
Setup and Availability	Laser Scanning	High precision—laser scanning offers measurements that are incredibly exact and precise, down to the millimetre High-quality imaging—laser scanning creates detailed 3D models and photos of the amputated limb
	Bioimpedance	Accurate—the measurements of residual limb volume using bioimpedance are incredibly accurate; the measurements may be verified by repeating them several times
	Water Displacement	Proper experiment setting is required because the patient’s position throughout the procedure may be uncomfortable or even harmful for some patients Minimally invasive procedure—the residual limb must be submerged in water to obtain the water displacement quantity; since the residual limb will be in direct touch with the water, there is a considerable risk of infection if the water is not so clean
	Anthropometric Measurement	Anthropometric measurements are a non-invasive way to determine the volume of the residual limb; it is less uncomfortable and hazardous than other conventional procedures because it does not involve the use of needles or incisions Simple to use—anthropometric measurements can be carried out quickly and easily without the need for specialised training or equipment
	Contact Probes	Contact probes are invasive because they need to come into direct contact with the residual limb, which some patients may find uncomfortable or painful Risk of infection—because contact probes have the potential to penetrate the skin, there is a risk of infection associated with their use Technical skill required—operators must be taught and have expertise to obtain precise measurements when using contact probes, which require technical skill to utilise appropriately Reduced operator variation—because contact probes give a reliable and objective measurement, they lower the possibility of operator variation
	Optical Scanning	Non-invasive—optical scanning is a painless, low-risk, non-invasive approach because it does not require contact with the residual limb Quick and easy to use—measurements of residual limb volume can be made quickly and easily using optical scanning, which only takes a few minutes to complete Skilled operators—requires professional and trained operators to produce precise measurements
	SXCT	SXCT is a minimally invasive method that is painless and has a low risk because it does not involve any touch with the residual limb Limited availability—not all healthcare facilities may have SXCT equipment readily available, which restricts certain patients’ access
	MRI	Non-invasive—MRI is a low-risk, painless procedure that does not require any contact with the residual limb Limited availability—not all healthcare facilities may have easy access to MRI equipment, which restricts certain patients’ access
	Ultrasound	Ultrasound is a painless, low-risk procedure that is non-invasive and does not involve any touch with the residual limb. Limited precision—ultrasound measurements might not offer as much precision as other techniques like contact probes or laser scanning Operator skill is needed—accurate measurements with ultrasound require a knowledgeable and trained operator Limited availability—not all clinics or healthcare facilities may have ultrasound equipment, which restricts certain patients’ access

Table 5. Cont.

Theme	Technique	Details
Application	Laser Scanning	Non-invasive—laser scanning is a painless, low-risk method that does not involve any touch with the residual limb Expert operators—accurate measurements from laser scanning require professional and trained operators
	Bioimpedance	Non-invasive—bioimpedance measurement is a non-invasive technique for calculating the volume of the residual limb; it is less uncomfortable and hazardous than other conventional procedures because it does not involve the use of needles or incisions Technical restrictions—in some patients, especially those with oedema or big limbs, bioimpedance measurements may be difficult to achieve; temperature fluctuations and the degree of skin hydration may also have an impact on measurement accuracy; and the measuring protocols need to be standardised to make sure that measurements are comparable and consistent across various medical professionals and devices
	Water Displacement	Water displacement is an objective way to measure the volume of the residual limb; this indicates that the measurements are reliable and unaffected by the operator's biases or prior knowledge Widely accepted—for measuring the volume of residual limbs, the water displacement technique has been in use for many years; safety and hygiene issues could arise because of this technique's use of water, so reduce the chance of infection, the water must be clean, and falls must be avoided by using a non-slip surface Patient discomfort—patients who have open wounds or ulcers on their residual limbs may find the water displacement treatment to be uncomfortable or bothersome Application restricted—soft tissue on the residual limb is only used for water displacement measurements; it does not evaluate the volume of the bone or muscle
	Anthropometric Measurement	The anthropometric approach has been used for many years and is generally accepted as a standard way to calculate residual limb volume Limited use—the volume of bone and muscle tissue in the residual limb cannot be accurately measured via anthropometric measurements Unreliable measurements—anthropometric measurements can differ greatly across different operators, and the application of various procedures might lead to unreliable outcomes Measurement accuracy is affected by external factors—anthropometric measurements might be inaccurate due to external factors like the patient's position or how tightly the measuring tape is wound
	Contact Probes	Ability to evaluate bone and muscle volume—patients who need a more thorough evaluation of residual limb volume may find it important that contact probes be utilised to evaluate bone and muscle volume
	Optical Scanning	Imaging—optical scanning can produce a 3D model or image of the amputated limb that can be used to help with the design and fitting of prosthetics The internal volume of the residual limb is not directly measured via optical scanning, which may result in inaccurate measurements if the scanning is not performed correctly
	SXCT	High-quality 3D models and photos of the residual limb are provided using SXCT's 3D imaging technology, which can be helpful for designing and fitting prosthetic limbs SXCT's ability to penetrate deeply into tissues allows it to measure volumes that are not achievable using other techniques like optical or laser scanning High radiation exposure—ionizing radiation is used in SXCT, which puts patients at risk for a high radiation dose

Table 5. Cont.

Theme	Technique	Details
	MRI	Can quantify deep tissues—MRI can penetrate deep tissues and quantify volume from places that may not be reachable by other means Ionizing radiation is not used in MRI; hence, there is no chance for patients to be exposed to it MRI has various limitations that must be taken into consideration; patients with metallic implants, for instance, might not be able to have an MRI, and those who have claustrophobia might find the procedure painful
	Ultrasound	Deep tissue measurement—because ultrasound can reach deeper into tissues, it is possible to estimate volume in places that may be difficult to access using other techniques Ultrasound only delivers two-dimensional images rather than three-dimensional ones, which makes it less effective for prosthetic design and fitting
	Laser Scanning	Rapid and simple to use—laser scanning is a rapid and simple technique that can evaluate residual limb volume in a matter of minutes Limited penetration depth—laser scanning’s accuracy may be constrained by its inability to penetrate deeper into tissues of larger thickness
	Bioimpedance	Objective—measuring the volume of the residual limb using bioimpedance is objective; this indicates that the measurements are reliable and unaffected by the operator’s biases or prior knowledge Bioimpedance measuring equipment is often lightweight and portable Limited use—only the soft tissue of the residual limb can be measured using bioimpedance; it does not evaluate the volume of the bone or muscle

Table 6 indicates the selected articles’ evaluation based on the reliability and validity of the residual limb volume measuring techniques. Based on the authors’ discoveries, most of the articles have conducted test–retest reliability assessments ($n = 29$), evaluated the reliability related to measurement conditions involving subjects ($n = 37$), and scrutinised the equipment’s reliability ($n = 20$). Following this, two types of validity tests were explored in this analysis, specifically criterion and construct validity. Among these, only 14 articles reported performing criterion validity tests, wherein the research compared the employed measurement techniques against the gold standard of measurement methods. None of the articles, however, incorporated the use of construct validity tests in their evaluations.

Table 6. Summary of the selected articles’ evaluation based on reliability and validity of residual limb volume measuring techniques. Order is based on publication year (from oldest to most recent year).

Studies	Reliability				Validity		
	Test–Retest Reliability	Operator Reliability		Equipment Reliability	Measurement Condition (Subjects) Reliability	Criterion Validity	Construct Validity
		Intra-Operator Reliability	Inter-Operator Reliability				
Fernie et al. [26]	UC	-	-	✓	✓	UC	-
Thomas W. Starr [27]	✓	-	-	UC	✓	-	-
Fernie et al. [43]	-	-	-	✓	UC	-	-
Golbranson et al. [21]	✓	-	-	✓	✓	✓	-
Krouskop et al. [28]	-	-	-	✓	✓	✓	-
Öberg et al. [44]	-	-	-	-	✓	-	-

Table 6. Cont.

Studies	Reliability				Validity		
	Test–Retest Reliability	Operator Reliability		Equipment Reliability	Measurement Condition (Subjects) Reliability	Criterion Validity	Construct Validity
		Intra-Operator Reliability	Inter-Operator Reliability				
Schreiner et al. [36]	UC	-	-	✓	UC	-	-
Smith et al. [29]	✓	✓	-	SXCT only	✓	✓	-
Vannier et al. [58]	✓	-	-	✓	✓	✓	-
He et al. [42]	-	-	-	✓	✓	-	-
Vannah et al. [34]	✓	-	-	UC	✓	✓	-
Commean et al. [1]	✓	✓	-	✓	✓	-	-
Johansson and Oberg [30]	✓	✓	✓	-	-	✓	-
Boonhong [31]	✓	-	-	-	✓	✓	-
Zachariah et al. [5]	✓	-	-	-	✓	-	-
Buis et al. [41]	UC	-	-	-	✓	-	-
Singh et al. [24]	-	-	-	-	UC	-	-
Sanders et al. [48]	-	-	-	✓	✓	-	-
McGarry et al. [35]	✓	UC	-	UC	✓	✓	-
Sanders and Lee, [38]	-	-	-	✓	✓	-	-
Sanders et al. [12]	✓	-	-	-	✓	-	-
De Boer-Wilzing et al. [32]	✓	✓	✓	✓	✓	✓	-
Sanders et al. [49]	-	-	-	-	✓	-	-
Sanders et al. [51]	-	-	-	-	✓	-	-
Sanders et al. [50]	✓	-	-	-	✓	-	-
Sanders et al. [52]	UC	-	-	-	✓	-	-
Tantua et al. [33]	✓	UC	-	UC	✓	✓	-
Sanders et al. [6]	✓	-	-	-	✓	-	-
J. E. Sanders et al. [53]	UC	-	-	✓	✓	-	-
Dickinson et al. [39]	✓	✓	✓	-	-	✓	-
Seminati et al. [45]	✓	✓	✓	-	✓	✓	-
Sanders et al. [54]	✓	-	-	-	✓	-	-
Kofman et al. [40]	✓	✓	-	✓	✓	✓	-
Sanders et al. [7]	✓	-	-	-	✓	-	-
Hinrichs et al. [55]	-	-	-	✓	✓	-	-
Youngblood et al. [8]	✓	-	-	-	✓	-	-
Larsen et al. [57]	✓	-	-	✓	✓	-	-
Youngblood et al. [56]	UC	-	-	-	✓	-	-
Paternò et al. [46]	✓	-	-	-	✓	-	-

UC = unclear/not discussed in detail.

4. Discussion

4.1. Techniques for Measuring the Changes in the Residual Lower Limb

4.1.1. Water Immersion

The water immersion measuring technique is one of the most frequently used techniques for determining the residual limb volume. Most publications used this technique as their 'gold standard' in performing validity tests [28,29,31,32]. In this technique, the amount of water displaced from a tank is measured and calculated as the residual limb volume (Archimedes' principle). The measurement is conducted by asking the subject to lower their residual limb into the tank until it reaches a specific marker [21,26,28–32], or water is pumped into the tank to immerse the residual limb [27]. During the measurement sessions, the knee should be in the position of the same degree of flexion/extension (knee flexed about 25°) throughout all the sessions, and the same marking placement for the residual limb immersion level should take place [21]. The temperature and atmospheric pressure of the water also need to be considered since they also affect the results [30]. This measuring technique is quite sensitive to the subject's movement as the subject needs to keep their residual limb at a constant position during the measurement sessions. Even one movement of the subject will produce inaccurate results [4]. Furthermore, since this is a contact measuring technique, it can distort the shape of the residual limb during the measurement sessions owing to the hydrostatic or buoyancy effects. Later, Smith et al. [37] made some improvements to this method by introducing a hydrostatic weighing technique that uses the apparent weight of water with residual limb immersion (i.e., there is no difference by measuring the amount of displaced water) as the residual limb volume.

4.1.2. Anthropometric Measurement

In the anthropometric measurement technique, the circumference of each segment of the residual limb volume is calculated using frustum and cylinder formulas and is summed up to determine the residual limb volume [31]. Normally, researchers use a normal measuring tape to measure the residual limb circumference, but there are some issues that may be related to the usage of a normal measuring tape. The first possible error that may occur during the measurement sessions depends on whether the researchers take the measurement in a consistent place and whether equal tension is applied while taking the measurement from one session to another [4]. Precautionary steps may be taken using appropriate tools such as the spring-loaded measurement tape [21,32,33], and only one researcher is allowed to take the circumference reading for all measurement sessions as they may apply a constant amount of tape tension. Another issue of this technique is in determining whether a 4 cm interval down to the distal of the residual limb [31,32], a 5 cm interval from the ischium level down to the residual limb tip [28], or one-eighth of an inch [21] is enough to determine the residual limb volume using frustum and cylinder formulas. Some validity tests may be needed to determine the best amount of interval and marking locations for the anthropometric measurement technique. This technique has been shown to be a poor technique for measuring the residual limb volume since the residual limb volume is not well related to the circumference, as reported by Golbranson et al. [21] and Tantua et al. [33].

4.1.3. Non-Contact Scanner (Laser and Optical Scanners)

Laser and optical scanners are residual limb volume measuring techniques that do not involve any contact with the residual limb since they are non-contact scanners. For the optical scanning method, a digital image is captured from different angles with multiple positions of the residual limb, and later, a 3D model image of the residual limb's surface is created with its calculated volume. There are two ways to produce optical scanning images, which are the silhouetting method [36] and fringe projection [29,32,38]. In the silhouetting method, the image is created from different angles of the outside contour of the residual limb, while for fringe projection, the grid pattern is projected onto the residual limb from different angles. For the laser scanner technique, a plane laser of light is projected

onto the residual limb from different angles with multiple locations, and video images are recorded. Three-dimensional images are constructed based on the recorded data to calculate the volume. Most of the time, non-contact scanners are used to capture the shape of the residual limb for digitalising the prosthetic socket design. Nevertheless, this application may also be used to determine the residual limb volume as the software can determine the volume of the scanned residual limb. Unlike the contact scanner, the optical scanning system captures radial measurements rather than circumferential measurements [4]. From this radial measurement, Geil [59] found that the better way to measure the distances on the residual limb is by using a calliper compared to using a non-contact scanner. Thus, a non-contact scanner has no benefit over circumference measurements in determining the residual limb volume change. One of the advantages of using a non-contact scanner is that it can capture the residual limb without applying any tension that may distort the residual limb's shape. It is quite easy to handle this non-contact scanner as it can adapt to limb movement during scanning [4]. Besides that, since it is a handheld scanner, it is portable and easy to carry everywhere. During the scanning process, the residual limb needs to be scanned as fast as possible since there is a high possibility for an increase or decrease in its volume as the change in the residual limb volume is very sensitive to socket doffing [5]. There are three publications on the validation of this measurement technique [30,32,33], and another three publications are on the validation of the scanner used in the studies conducted by Fernie et al. [43], Seminati et al. [45], and Kofman et al. [40]. Some scanners such as VIUScan, Go!SCAN, Artec Eva, Bioscaptor, and CAPOD demonstrated a high consistency of accuracy to determine the residual limb volume. However, more publications are needed in the validity test of laser scanning since several laser scanning brands in the market may show different results due to their own errors that sometimes affect the scanning results. Other than that, the proper alignment of shapes from different scanners may also be an issue in this technique. Some researchers placed fiducial markers on the residual limb to ensure the right alignment, and in commercial software packages, the residual limbs that have bony prominences or other regions of sharp curvature tend to align better than those without distinct features [1,29].

4.1.4. Bioimpedance

There is quite a high number of publications on bioimpedance measuring techniques. This technique is among the best technique that can be applied to measure the residual limb volume fluctuation due to its sensitivity. In the early stage of bioimpedance analyser instrument development, Sanders, Rogers, and Abrahamson [48] used a commercial bioimpedance analyser (Xitron Hydra 4200) that can measure impedance at 50 frequencies between 5 kHz and 1 MHz to determine the fluid resistance. The data collected were then used to estimate the extracellular fluid volume. The researchers used four strip electrodes, namely the outer pair (injected current) and the inner pair (sensed voltage) to collect the impedance data. Then, Sander et al. [50] enhanced the design of their device by changing the dimensions of the current injecting electrodes (proximal: 15×2 cm; distal: 3.5 cm diameter) and voltage sensing electrodes (7.5×2 cm). Other than that, previous studies customised the electrodes using conductive tape (0.09 mm thickness), custom multi-stranded silver-plated copper wire, and polyvinyl chloride (PVC) insulation. The wire was attached to the electrode by splaying its ends and sandwiching it between two pieces of conductive tape (the underside of the conductive tape was covered with hydrogel). A uniform-thickness hydrogel patch was also used in most studies. Meanwhile, the outside of the conductive tape was covered with Tegaderm. By using this measuring technique, researchers may determine the residual limb volume gain or loss while the residual limb is in the prosthetics socket. Thus, the analysis of the residual limb volume fluctuation in a special region can be conducted in real time. This technique can also be simultaneously conducted while the amputee is sitting, standing, or walking. Moreover, with this technique, researchers may also determine the possible causes of residual limb volume fluctuation, such as the health status of the subject [12], the prosthetic socket suspension

system [49,56], the socket design or socket size [52,57], activity level [6–8,50], and the post-doffing effect [51]. Miyatani et al. [47] also reported that this bioimpedance technique is highly correlated ($r > 0.9$) with the extracellular fluid volume measured using the MRI technique. However, the publication was only limited to the level of transtibial amputation. Hence, more research needs to be conducted on the transfemoral amputation level since there is more rapid volume loss from the huge number of soft tissue areas compared to transtibial amputation. The use of bioimpedance on transfemoral amputees also needs to be validated before it is implemented clinically.

4.1.5. Other Techniques (Contact Probes, Ultrasound, SXCT, and MRI)

There is a low number of publications on contact probe, ultrasound, SXCT, and magnetic resonance imaging (MRI) techniques. In the contact probe technique, the residual limb volume is determined by measuring the position of the small contact probe that touches the surface of the residual limb. The position measurement of the contact probe is conducted using an instrument that is built up with a mechanical arm in which the stylus arm rotates around the residual limb [28]. Alternatively, this technique can be conducted using an electromagnetic sensor [34,35]. For the ultrasound, SXCT, and MRI techniques, the residual limb volume is determined by constructing multiple 3D images that were captured. These techniques can construct both the internal and surface structures of the residual limb [4]. In the ultrasound technique, the image of the residual limb is captured by placing the residual limb inside a water-filled tank, and a non-contact ultrasound sensor moves around inside the tank. The captured image is used to construct the 3D image, and the residual limb volume can then be determined [42,60]. There is only one study, conducted by Singh et al. [24], that determined the residual limb volume using this ultrasound technique. Next, SXCT is the measurement technique that measures the residual limb while it is in the prosthetic socket [1], a technique which is like the bioimpedance technique. This technique has been also proven by Smith et al. [29] as well as by Smith et al. [37] to have less than 1.0% of volume errors. Other than that, there are only two publications that directly measured the residual limb volume using the MRI technique, namely Buis et al. [41] and Miyatani et al. [47], with one further study conducted by Safari et al. [61] that did not directly involve the residual limb. In terms of the time taken to finish the scan process, the ultrasound technique takes 780 s (approx. 13 min), and the MRI technique takes 592 s (approx. 10 min), which are more time-consuming compared to the SXCT technique, which only takes 32 s [37,58]. A technique with a long time of measuring will have a high potential of error due to subject movements.

4.2. Selection of the Appropriate Techniques According to Measurement Purposes

It is important to choose the right technique for measurement according to our needs and the purpose of measuring. In general, clinicians or researchers may measure the residual lower limb volume changes from macroscopic and microscopic aspects, and different techniques of measuring may be applied according to the specific post-operative period. The American Academy of Orthotists and Prosthetists (AAOP) State of the Science Conference on Post-Operative Management of the Lower Extremity Amputee stated that there are five main stages in the post-operative recovery period: (1) the pre-operative stage; (2) the acute hospital post-operative phase (5–14 days post amputation); (3) the immediate post-acute hospital stage (4–8 weeks post amputation); (4) the intermediate recovery stage (4–6 months post amputation); and (5) the transition to a stable stage (12–18 months post amputation).

At the early stage post amputation (phase 2), the residual limb normally experiences oedema, and this condition reduces over time. During this phase, it is important to monitor the shape and volume of the residual limb as a preparation for the first prosthetic fitting. The most suitable technique that may be applied to measure the residual limb volume during this phase is the anthropometric measurement technique since this technique is the simplest, fastest, and easiest way to measure the change in the residual limb. In

clinical practice, clinicians or prosthetists can easily record, keep, and compare the residual limb circumference measurement to determine whether the residual limbs are ready for the first prosthetic fitting. Then, during the first prosthesis usage (phases 3 and 4), the anthropometric measurement technique is no longer suitable for measuring the changes in the residual limb volume because the changes are less compared to the early post-amputation phase because of a decrease in oedema over time during phase 2. More accurate and precise residual limb volume measurement techniques are thus needed, since some publications found that the circumference measurement is not so related to the residual limb volume measurement. Water displacement, contact probes, ultrasound, MRI, and optical or laser scanning were seen to yield good outcomes to determine the changes in the volume during this phase. In phases 3 and 4, there are still changes in the residual limb volume that cause the residual limb to be unable to reach a stable state. Thus, correct volume measurement techniques help to determine whether the residual limb has already reached its stable state. When the residual limb has reached its stable state, only very small changes in the residual limb volume can be detected. Therefore, more sensitive measuring techniques such as bioimpedance and SXCT are required to study the changes in the residual limb volume. At the same time, these techniques are also able to measure the residual limb volume without worrying about the post-doffing effect. Although the SXCT technique seems to have benefits in this phase, the technique is not so suitable in all cases of measurement since it is only limited to the stationary position of the subjects. Only the bioimpedance technique can measure the volume changes in the residual limb while the subjects are standing and walking. Bioimpedance seems to have benefits in measuring the residual limb volume fluctuation in real time as researchers may directly analyse the effects of the suspension system, socket design, and activity level. Hence, researchers can not only concentrate on the residual limb, but can also pinpoint the precise location of volume gain or loss within a particular region.

Clinicians or researchers also need to understand the aim of measuring the fluctuation in the residual lower limb volume because the residual lower limb volume can be measured in two ways: (1) by measuring the residual limb volume as a whole and (2) by only measuring the change in the residual limb volume (i.e., neglecting the actual residual limb volume). Seven techniques can be used to measure the residual lower limb volume, and only the bioimpedance and SCXT techniques can be used to measure the change alone, i.e., the fluctuation in the residual limb volume. There are pros and cons while measuring the residual limb as a whole volume as opposed to just measuring the changes in the residual limb volume. When the residual limb measurement is conducted as the whole volume, clinicians can record and compare the volume measurement from time to time (from phase 2 to phase 5 post operation). It is a good practice in the clinical field when clinicians can track any changes in the residual limb volume as they can plan further for the amputee's improvement programme, especially in terms of prosthetic usage. At some point after the amputee has reached a stable state of their residual limb (phase 5 post operation), the clinicians can prescribe a better prosthesis component for the improvement of the amputee's quality of life. On the other hand, by just measuring the changes in the residual limb volume, there are a few advantages such as its speed in determining the residual limb volume fluctuation since it may directly give the final reading of fluctuation. Also, it is prone to have a sensitive measurement since it is not influenced by the post-doffing effects. Finally, another advantage is that it is a real-time volume measurement that is taken while the subject is walking or standing. In terms of the disadvantages of measuring the residual limb as a whole volume, researchers and clinicians are facing some difficulties in obtaining accurate results of the residual limb volume fluctuation. In most of the previous works, the researchers compared the residual limb volume before and after to obtain the amount of residual limb volume fluctuation. This means that the residual limb volume was not measured in real time. Some factors such as the post-doffing effect and the activity level of the amputee while performing the experiment possibly contributed to the wrong interpretations of the residual limb volume fluctuation. Even though the bioimpedance

technique could give good results of the volume fluctuation, this measurement technique is not so user-friendly compared to other techniques. The bioimpedance technique is difficult to use in clinical fields since it has some complicated setups that need to be run by skilful researchers. In addition, due to some limitations in the bioimpedance technique, there are only publications on transtibial amputation levels.

On the other hand, the laser scanning technique appears to be more user-friendly for both clinicians and prosthetic users as this technique can immediately be used at the clinic. The procedure of this laser scanning is simple to learn, and the scanning of the residual limb can be performed in a short time. Due to these reasons, this technique has potential for use in the clinical field. However, some instrumentation errors need to be resolved, and perhaps more importantly, the results must be proven to be accurate. Other than these, the interaction of the laser with soft tissue and numerous other issues vary by product. Thus, these variations and issues need to be resolved and improved. Additionally, the laser scanning technique also has some limitations because the residual limb volume may be impacted by the post-doffing effect since the prosthetic socket must be doffed before scanning can be completed, and slow scanning times will change the residual limb volume.

As previously discussed, the authors concluded that choosing the appropriate technique is crucial for evaluating the residual limb volume. Since the residual limb can be quantified from macroscopic and microscopic aspects, researchers or medical professionals must properly select the appropriate technique to evaluate the residual limb volume. Clinicians or researchers may employ anthropometric measurement techniques to ascertain the residual limb volume change from the macroscopic perspective, particularly in phase 2 of post-operative recovery where there are still significant residual limb volume changes. Then, more precise measurement techniques, including water displacement, contact probes, optical scanning, MRI, ultrasound, and laser scanning, can be used during the other post-operative periods (phases 3, 4, and 5). On the other hand, the bioimpedance and SXCT techniques are available to researchers that want extremely sensitive measurements, but these two techniques can only quantify the change in the residual limb without knowing the total volume of the residual limb. In terms of the application of measuring techniques in the clinical field, the clinician needs to see the availability of the measuring device in their clinical setting. Since some techniques such as anthropometric measurement and laser scanning use a portable instrument, these techniques are seen as easier to use in clinics compared to other approaches. Even MRI, ultrasound, and SXCT devices are accessible in most hospital setups, but the costs associated with these procedures are significant, and other techniques like bioimpedance, contact probes, optical scanning, and water displacement require specific experiment setups that are challenging in a clinical setting. With the right selection of measuring techniques, the researcher or clinician will be able to evaluate the residual limb volume in line with their aims and interests.

4.3. Sources of Error and Precautionary Steps

There are four potential sources of error associated with the residual limb volume measurement procedure for the lower limb. First, the final volume measurement is extremely sensitive to subject or residual limb movement in a few measurement techniques including water displacement, contact probes, laser scanning, MRI, and ultrasound. Subject movement can be caused by several factors including a lengthy scanning time, the subject's inability to stand for an extended period, and the subject's inability to stand still. Second, when the researcher employs certain measuring procedures, there are errors due to the distortion of the residual limb's shape that happens when force is applied to the skin and creates a compression of the residual limb. These errors are presented by the anthropometric measurement, contact probes, and water displacement techniques. Third, the post-doffing effect also contributes directly to the source of error. All techniques except for SXCT and bioimpedance are susceptible to this error. After removing the prosthesis, the residual lower limb volume will have some changes in its volume. Slow-capturing residual limb volume measuring techniques will result in varying volume estimations. To minimise

the misinterpretation of volume readings, researchers need timing criteria for measuring the residual limb volume of the lower extremity. Lastly, device resolution and precision may also potentially lead to errors in measuring the residual lower limb volume. This type of error is caused by faulty equipment used by the researchers to measure the volume of the residual lower limb.

The potential errors in various techniques may contribute to different interpretations of the residual limb volume even though researchers measure the identical residual limb subject. It is challenging to obtain the same residual limb volume across studies when utilising various measuring techniques, but to properly interpret the residual limb volume, researchers must evaluate whether the measurement technique's error was greater than the changes in the residual limb volume. It is a good research practise for researchers to compare or use many techniques in studies in order to identify the best approaches with the least amount of error, as demonstrated by de Boer-Wilzing et al. [32], Tantua et al. [33], Johansson et al. [30], Golbranson et al. [21], Krouskop et al. [28], Smith et al. [37], and Boonhong [31]. In previous studies, it is hypothesised that water displacement is the most accurate measuring technique [28,29,31,32], but as other techniques have been developed, this water displacement technique cannot be set as the only gold standard in evaluating the other techniques. The water displacement technique is thought to have a large potential for error, especially when it involves repetitive sessions, since the researchers must ensure that the residual limb has the same level of immersion and knee flexion, and that the person cannot be moved while the measurement is being taken. In addition, as this technique is a contact measurement type, extra caution should be taken because surface tension between the residual limb and the water during immersion has a large potential for error. With the developed technology, laser scanner techniques are potentially seen to be more accurate in giving the volume of the residual limb compared to other techniques. Although the laser scanning approach has several limitations, the authors still consider it to be an effective measurement technique with less potential error because it is a fast-measuring technique. As stated by the authors, one of the issues with this laser scanning technique is related to subject movement rather than the interaction of the laser with soft tissue. Additionally, subject movement may contribute to issues in achieving the proper alignment of shapes from different scans. Even though this technique has several limitations, it has the advantages of minimising subject movement, limiting the post-doffing effect, and preventing the distortion of the residual limb's shape because there is no contact with the residual limb while measuring. The digitalisation of this laser scanning technology has also resulted in improved error reduction because researchers can collect and analyse data in the CAD software (TT Design 5.0, 2010; Geomagic Studio 2012.1.1, 2012; VXElements 6.1, 2017; Canfit™ 15, 2019; Artec Studio Software 16, 2020) after scanning. Researchers can use the CAD software to compare the images obtained from various types of scanners, which allows them to determine which scanner is the best to use. Johansson et al. [30], Seminati et al. [45], and Kofman et al. [40] conducted studies to compare the performance of various types of scanners and discovered that the majority of scanners (Artec Eva scanner [45], TT Design, Omega Scanner, BioSculptor Bioscanner, Radio4D Scanner [40], and CAPOD [30]) are reliable in determining the residual limb volume. Aside from that, optical scanning techniques (both silhouetting and fringe projection) also demonstrate the ability to capture images of residual limbs very quickly. The ability of this technique to capture the image in less than 1.5 s was proven by studies conducted by Vannier et al. [58] (0.75 s), Schreiner and Sanders (1.1 s) [36], and Sanders and Lee (1.5 s) [38]. Even the optical technique can capture the image of the residual limb very quickly, but it still has a significant chance of error since the silhouetting technique cannot capture the concavities of the residual limb. The subject's position during capture is also considered a potential source of error for the fringe technique since the subject must be placed so that the residual limb may be viewed from all angles. The researchers will have some difficulty capturing the image and will also come up with a wrong interpretation of the residual limb volume if the individuals have certain constraints in positioning their bodies in a proper position, particularly for transfemoral amputation

and elderly subjects. The authors of the review also found that bioimpedance techniques appear to have fewer sources of error; however, this technique cannot be demonstrated to be 100% accurate because it only evaluates the residual limb volume change and cannot be compared to other techniques. As was previously mentioned, the bioimpedance and SXCT techniques both only assess the volume change in the residual limb, but it is difficult to compare the two techniques because the SXCT technique can only measure the residual limb in stationary positions. More research on the reliability and validity of the bioimpedance and SXCT techniques is required to demonstrate that they are the most accurate and have the fewest potential errors because the authors were unable to locate any studies relating to these topics, and most of the studies that were conducted only concentrate on the transtibial amputation level. Other techniques such as the anthropometric measurement technique also show their weaknesses since they have a high chance of error. The placement of the measurement device while measuring the residual limb circumference can be viewed as a potential source of error since the researchers need to position the measurement tape at the same place and, at the same time, apply the same amount of tension during the measurement to avoid any error. In the study conducted by Tantua et al. [33], it was also discovered that the circumference measurement was not related to the residual limb volume, which, at the same time, will give the wrong interpretation of the residual limb volume measurement. Finally, because the post-doffing effect directly affects the residual limb volume, the lengthy duration of measuring techniques may potentially contribute to inaccuracies in the interpretation of the residual limb volume. Ultrasound and MRI techniques have the potential to somehow be included in this error because both are very time-consuming, taking between 592 s [41] and 780 s [42,60] to complete the measurement. Like most of the techniques, both techniques are also very sensitive to the subjects' movement, which will later cause errors in the final residual limb volume reading. Researchers should take extra precautions to avoid evaluating the residual limb volume incorrectly. Precautionary steps need to be taken while experimenting, especially when it involves repeated session tests. Researchers should monitor the subjects' activity using activity monitoring devices such as those utilised by Sanders et al. [54], as different amounts of activity levels may directly affect the results. Activity monitoring devices may monitor the activity to give a similar amount of activity levels. For example, in between morning and afternoon sessions, the subjects are allowed to have their own activity without being monitored. For some subjects, they can just sit down outside of the lab for three hours, while on another day, they perform various activities. These situations may cause inaccurate results of the residual limb volume changes and directly give the wrong interpretation of the residual limb volume fluctuation. Other than that, the marker placement or position during the experiment is also important throughout all the experiment sessions. Different marker positions, especially for immersion indication in the water displacement technique, would cause different amounts of water to be displaced from the tank in every different measurement session. It is good to use an ink marker that cannot be rubbed off and displaced throughout the sessions. Some techniques such as water displacement, MRI scanning, and SXCT technology necessitate the subjects or residual limb not to move until the measurement session is complete. To minimise erroneous interpretations of the residual lower limb volume in the subjects who cannot stand still or who do not move during the measurement sessions, different procedures are preferable. Furthermore, all nine techniques except for SXCT and bioimpedance can measure the residual limb outside of the prosthetic socket. Precautionary steps need to be taken as the residual limb volume will affect the post-doffing volume change effect. Fast capture techniques such as bioimpedance, SXCT, and optical or laser scanning need to be used to investigate the post-doffing effect as opposed to the slow measuring techniques such as MRI, ultrasound, contact probes, and water displacement.

In summary, all measurement techniques can lead to errors. There are four potential sources of error associated with the residual limb volume measurement techniques for the lower limb, which include (1) subject or residual limb movement, (2) the distortion of the

residual limb's shape, (3) the post-doffing effect, and (4) the resolution and precision of the used device. Also, according to the authors' findings, there are three main causes of this sort of error, which are (1) the individual performing the measuring, (2) the subject being measured, and (3) the measuring device itself. Researchers can minimise potential errors while measuring the residual limb volume by (1) monitoring the subjects' activity during the measurement session, (2) using high-quality markers that cannot be easily rubbed off and avoiding different marker placements, and (3) choosing the right techniques according to the aim of the research and avoiding using measuring techniques that require a lengthy time to reduce subject movement. Correct precautionary steps may reduce errors while measuring and lead to a more accurate interpretation of the residual limb volume.

4.4. Improvement of Measuring Techniques for the Past 10 Years

As discussed earlier, one of the aims of this systematic review is to analyse the improvement of the techniques used for measuring the residual limb volume, and this study acts as an extension of the study by Sanders and Fatone [4]. Since the last publication in the year 2011, a lot of researchers have shown their interest in measuring the residual limb volume. There are 21 publications from the past 10 years related to the techniques for measuring the residual limb, and a lot of improvements were conducted by researchers to make the techniques more accurate, user-friendly, and easy to apply. Apart from that, for the past 10 years, based on the published articles, researchers have presented their methods and data in a clear and easy-to-understand way to allow for any replications of the techniques. This systematic review also found that there is an increase in the number of studies in both the laser scanning and bioimpedance techniques.

In recent years, the bioimpedance technique has shown an improvement. Some researchers made the technique easy to use and more sensitive to any changes in the residual limb. There are 12 publications on using the bioimpedance technique to measure the residual limb from the past 10 years, and this number of publications shows a drastic increase compared to the number of publications before 10 years ago, when there were only 2 publications related to this measuring technique. Researchers conducted some modifications to the techniques, especially in relation with the electrodes, as they can suit the residual limb and, at the same time, make the electrodes more sensitive to any volume measurement changes. Sanders et al. [53] improved and developed their own electrode that can suit the usage of bioimpedance instruments in the residual limb volume measurement studies. The developed electrode was thin with a thickness of 0.09 mm, flat, and could withstand mechanical and shear stresses inside the prosthetic socket. These features can help give good results since the electrode will not disturb the normal prosthetic socket configuration. Moreover, compared to the electrode used by Sander et al. [48] before 10 years ago, they just used the commercial bioimpedance instruments that had a thickness of 0.81 mm, which led to less accuracy in the signal reading. Another study also discovered that a thin layer of hydrogel ensured a good electrical coupling [53]. Later, in the year 2019, Hinrichs et al. [55] developed a portable bioimpedance instrument that helped researchers to monitor the residual limb volume changes outside of the lab setting. With the development of the bioimpedance instrument, the results were more accurate and reliable, showing that the instrument can be used to measure the volume changes inside or outside of laboratory settings. As mentioned earlier, this technique can measure the volume changes in real time only without knowing the residual limb volume. Thus, this allowed the researchers to measure the residual limb volume changes in every step of the gait cycle. This technique can determine the possible causes of residual limb volume changes without worrying about the post-doffing effect. As reported by Sander et al. [6,49–51,54] and Youngblood et al. [56], some factors such as the type of socket suspension system (vacuum, non-vacuum [49], and suction socket [56]); the post-doffing effect [51]; the amount of residual limb volume changes within one day [50]; daily activity such as walking, standing, and resting [6]; and the effect of the socket size (oversize, normal, and smaller sizes) [54] were found to be the causes of volume changes in the residual limb.

Apart from the bioimpedance technique, the laser scanning method also shows an increased trend since the past 10 years. There are six publications related to the laser scanning technique from the years 2011 to 2021 as opposed to only three publications before the past 10 years. With the development of the computer-aided design and computer-aided manufacturing (CAD-CAM) industry, the laser scanning technique has become popular as it is simple and easy to use. In line with this development, various types of scanners were introduced, and there are many studies that tested the accuracy and validity of the scanners. Some studies [32,33,39,40,45] have proven that this measuring technique is valid and reliable in determining the residual limb volume. The handheld scanners such as TT Design, Omega Scanner, BioSculptor Bioscanner, Radin4D Scanner, Artec Eva scanner, VIUScan marker-assisted laser scanner, Go!SCAN 3D structured white scanner, and Sense 3D markerless laser scanner that were used in the studies have shown small mean percentage errors (less than 2%) [45]. This laser scanning technique was also compared with other measuring techniques such as the water displacement, anthropometric measurement, and optical scanning techniques [32,33], and these studies discovered that the laser scanning technique had the lowest repeatability coefficient [32], which indicates that this technique has the best reliability. The laser scanning technique is also user-friendly as it does not involve any contact with the residual limb while the scanning process is conducted. Also, it takes less time to scan the limb (60–240 s) [39], and the scanning method is easy to learn (based on Post-Study System Usability Questionnaire) [40]. Comparing these with the years before 10 years ago, fewer studies focused on this technique. During that time, there was no usage of a handheld scanner in the research, and the residual limb measuring apparatus was complicated and not portable [30,43,44]. The measuring device may have caused difficulty to the subjects as they had to sit or stand on the measuring device, and the measurement results may have been affected by subject movement as well as the post-doffing effect, which would give a wrong interpretation of the measured volume as the subjects needed some time to stand or sit on the measuring device. Even though these techniques (during that time) applied quite a large and unportable measuring apparatus, studies still showed that this technique (CAPOD System) had lower random and systematic errors compared to the optical scanning technique (ShapeMaker) [30]. Thus, the findings prove that the laser scanning technique was reliable to use and gave good results.

Other techniques seem to be less popular for the past 10 years' publications, with only one article published for water displacement, anthropometric measurement, and optical scanning, while no publication was found for other techniques (contact probes, SXCT, MRI, and ultrasound). Some factors that may contribute to these unpopular techniques include less sensitivity of the devices used in the techniques, less accurate results due to the post-doffing effect, difficult positions of the subjects, techniques that involve contact with the residual limb until they cause distortion to the residual limb, cost, and complicated setups.

4.5. Applications of the Measuring Techniques in the Clinical Field

This systematic review also aims to provide clinicians with a better understanding of how to execute the residual limb volume measurement and, at the same time, how to apply it to clinical fields. Some measurement techniques have the potential to be used for this purpose. Since some measuring techniques have been improved in their ways to perform the tests, measuring techniques seem to have become easier to learn and handle, and some techniques are user-friendly, with some portable devices used for measuring. The improvement of the measuring techniques potentially allows for clinicians to monitor their patients' residual limb volume in the clinical field as this is important for preparing them for prosthetic fitting or monitoring the residual limb volume in cases when a prosthetic socket change is needed.

Currently, in most clinics, the way of recording the residual limb volume is by measuring the residual limb circumference at a certain point. This technique of recording is not a proper way of recording since the residual limb volume does not correlate well

with circumference [33]. The laser scanning technique is one of the best techniques that can potentially be used to record the residual limb volume in the clinical field since this technique is user-friendly, easy for a clinician to learn, and has a short scanning time, and all the data can be saved digitally. These are unlike other techniques such as the bioimpedance technique, which is not so suitable to use for clinical applications since it can only record the amount of change in the residual limb volume. In addition, the techniques such as water displacement, contact probe, optical scanning, and bioimpedance have difficult setups for measuring the residual limb volume, while other techniques such as SXCT, MRI, and ultrasound are costly.

5. Conclusions

One of the purposes of this systematic review was to examine the current improvements in the techniques available for measuring the residual lower limb volume (especially in the past 10 years). Other than that, this review also aimed to provide a better understanding to all interested parties, especially clinicians, on how to execute the measurement of the residual limb volume and will serve as a reference source for the development of more sophisticated residual limb volume measurement technology in the future. This systematic review identified 904 potential articles, and only 39 articles were selected to be reviewed based on the stated criteria. From the findings, there were nine techniques that were identified for measuring the residual limb volume, namely water displacement, anthropometric measurement, contact probes, ultrasound, spiral X-ray computed tomography (SXCT), magnetic resonance imaging (MRI), optical scanning, laser scanning, and bioimpedance. All the techniques were assessed and analysed by the authors. Generally, the residual lower limb volume can be measured in two ways: (1) by measuring the residual limb as the total volume and (2) by simply measuring the changes in the residual limb volume alone (i.e., neglecting the actual residual limb volume). Bioimpedance was found to have the best outcome since this technique has a high level of sensitivity and can measure the in-socket residual limb volume fluctuation. However, the publications were limited to only transtibial amputation. As an alternative, the laser scanning technique may be used to determine the residual limb volume for both transfemoral and transtibial amputations for the post-doffing volume measurement. Handheld scanners that are normally used in laser scanning techniques are easy to handle, and thus, fast scanning may be performed in a way to reduce the post-doffing effect. Subject movement, the distortion of the residual limb's shape, the post-doffing effect, as well as the accuracy and resolution of the measurement device are the four main sources of error that need to be considered during the measurement sessions. Furthermore, precautionary steps need to be taken to avoid obtaining wrong interpretations of the residual limb volume. By identifying the potential errors and precautionary measures needed in the techniques, researchers can improve future technology, potentially making it easier, faster, and error-free. The correct determination of the residual limb volume fluctuation would help to improve the prosthesis quality, especially the prosthetic sockets, and improve the amputee's quality of life at the same time.

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Conflicts of Interest: The authors declare no conflict of interest.

Appendix A

Search Syntax	
Search Date: March 2021–December 2021	
Database Search	Search terms in databases
PubMed In title, abstract, keywords	residual limb OR lower limb OR transtibial OR transfemoral OR amputee OR amputated OR amputation OR above knee amputation OR below knee amputation OR
ScienceDirect In title, abstract, keywords	fluctuation OR volume OR volumes OR circumference OR
Web of Science In title, abstract, keywords	water displacement OR anthropometric OR contact probes OR optical scanning OR spiral OR x-ray computed tomography OR magnetic OR resonance imaging OR ultrasound OR laser scanning OR bioimpedance OR CAD/CAM AND
Google Scholar In title, abstract, keywords	technology OR technologies OR technique OR techniques OR measurement OR measurements
Search Limit: 1975 to 2022	

Appendix B

Quality Assessment

Reference: _____

Assessment of Subject Selection/Details					
#	Questions	Y	N	U	Remarks
1	Are the number of subjects, type of amputation, or any objects used reported?				
2	Time since amputation/phase of post amputation—Do the subjects have the same phase of amputation?				
3	Comorbidities of subjects—Do the subjects have the same health level status?				
4	General activity level—Do the tested subjects have the same activity level as the other participants or are they able to stand or walk independently?				
Assessment of Methodology—Does the methodology have enough details to allow for replications?					
#	Questions	Y	N	U	Remarks
5	Is the position of the residual limb/subject during the test reported?				
6	Is the marker/landmark used to calculate the residual limb volume stated? Clearly state the position of the marker or where to put the marker or from which landmark the volume was calculated				
7	Is any time or delay reported to take the measurement? Can cause post-doffing effect				
8	Any repeated sessions?				
Assessment of Reference Standard					
#	Questions	Y	N	U	Remarks
9	Used more than 1 technique—validity test				
10	Does the result compare the measurement with the most accurate reading, e.g., set the water displacement results as the 'gold standard' to validate other techniques or use any known volume objects and compare the results with techniques used				
Assessment of Results					
#	Questions	Y	N	U	Remarks
11	Were all measurement results clearly reported?				
12	Explain how to record the results, e.g., calculation, comparison technique				
13	Were any statistical results reported (including simple statistical analysis), e.g., mean, SD, validity, or reliability				
Score Y = Yes; N = No; U = Unclear.					
#	Number of questions				

Appendix C

Study evaluation based on reliability and validity theme.

Reliability		
#	Evaluation Criteria	Explanation
1.	Test–Retest Reliability	The researchers will conduct multiple measurements on the identical subjects at various time points to assess the dependability of the techniques employed for measuring fluctuations in volume of the residual lower extremities. Normally, to ascertain the stability of these measurements across time, the intraclass correlation coefficient (ICC) is computed.
2.	Intra-Operator Reliability	Within the study, the measurements could potentially be taken by multiple researchers or operators. Compute the ICC to assess the reliability and consistency of measurements conducted by the same operator at different instances. or The goal is to gauge how reliably an individual performs measurements on a residual limb.
3.	Inter-Operator Reliability	The identical residual limb might be measured by various operators or researchers. Calculate the ICC to determine the reliability of measurements carried out by multiple operators. or This pertains to the consistency exhibited by different individuals when measuring the identical residual limb.
4.	Equipment Reliability	Guarantee the precision and upkeep of the instruments, equipment, or devices utilised for measurement techniques. Conducting repeated measurements on the same limb should result in dependable data from the equipment. Regarding scanning techniques: To attain uniformity in measurements across different scanning sessions, researchers must uphold consistent scanning parameters, including aspects like lighting, positioning, and calibration.
5.	Measurement Condition (Subjects) Reliability	For consistent measurements across multiple sessions, researchers need to establish highly precise measurement conditions when assessing the residual limb. This involves maintaining unwavering measurement parameters, such as proper skin preparation, electrode implantation, and consistent body positioning, to ensure reliable measurements across different sessions.
Validity		
#	Evaluation Criteria	Explanation
6.	Criterion Validity	Contrast the utilised techniques against a gold standard/benchmark method, like water displacement, MRI, or CT scan, known for delivering accurate measurements of limb volume. To gauge the degree of similarity between measurements acquired through various techniques and the gold standard, calculate correlations, and create Bland–Altman plots.
7.	Construct Validity	Examine the underlying theoretical principles of the employed measurement techniques for assessing fluctuations in limb volume. Verify whether the procedure aligns with the expected physiological changes that residual limbs typically undergo over time.
#	Number of questions	

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